

Mahmudul Hoque

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TECHNICAL SKILLS

Programming: Python, SQL, C, MATLAB, Bash

Machine Learning: Deep Learning, Computer Vision, Large Language Models (LLMs), Vision-Language Models (VLMs), Multimodal Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Generative Models (GANs, Diffusion Models), Parameter-Efficient Fine-Tuning, Model Quantization, Distributed Training, Synthetic Data Generation, Object Detection, Segmentation, Clustering

Frameworks & Libraries: PyTorch, Hugging Face Transformers, PEFT, Accelerate, Diffusers, BitsAndBytes, FAISS, FastAPI, MONAI, OpenCV, scikit-learn, scikit-image, NumPy, SciPy, pandas, Matplotlib

Platforms & Tools: Google Cloud Platform, High-Performance Computing (SLURM), Linux/WSL, Git, Docker, Conda, Visual Studio Code, Weights & Biases, Tableau

EXPERIENCE

Machine Learning Researcher

Aug. 2023 – Present

Morgan State University

Baltimore, MD

- Built a model-intrinsic explainability system for radiology image captioning using 0.5B–72B LLaVA-OneVision models; **ranked 1st** in the radiologist-evaluated **ImageCLEFmedical 2026 Explainability Challenge**, leading 6 of 9 criteria and earning an overall score 30% higher than the runner-up.
- Extracted decoder attention over 729 image patches to generate word-level heatmaps and bounding boxes, showing which image regions the model attended to for each generated caption word.
- Engineered fault-tolerant training and inference pipelines on A100/H200 multi-GPU clusters using PyTorch DDP and SLURM, enabling QLoRA/QDoRA fine-tuning of a 72B VLM on 116,604 images and inference on 15,249 test images.
- Benchmarked six model-adaptor configurations, measuring $5.7\times$ the inference latency at 72B versus 0.5B scale and peak GPU memory of 2.87–84.41 GB; released reproducible adaptors on **Hugging Face**.
- Developed and fine-tuned multimodal radiology captioning models using LLaVA, IDEFICS-9B, and VisionGPT2; **placed 2nd** in the **ImageCLEFmedical 2024 Caption Prediction Challenge**, earning the highest MedBERTScore and a BERTScore within 0.3% of the best-performing team on that metric.

Research Associate

Jun. 2024 – Jun. 2025

National Institute of Standards and Technology (NIST)

Gaithersburg, MD

- Developed a 3D-GAN model to synthesize additive-manufacturing defects, adding a novel loss module that reduced fragmented outputs; over 98% of generated samples kept at least 95% of pore volume in a single connected component.
- Created and published a **dataset of 3,503 real and synthetic 3D pore volumes** with morphological measurements; registration-based similarity (median IoU < 0.70) confirmed synthetic pores were structurally novel.
- Seeded size-stratified synthetic pores into a 3D volumetric model of an additive-manufacturing bracket, generating simulated XCT volumes with known defect ground truth for detection and nondestructive-evaluation benchmarking.

Data Analyst Intern

May 2023 – Aug. 2023

iFarmer Limited

Dhaka, Bangladesh

- Developed an automated National ID information-extraction pipeline using YOLOv8 for field detection and PaddleOCR for text recognition, achieving 93% accuracy on date-of-birth and ID-number extraction.
- Fine-tuned Llama 2 on agriculture-domain text to build a domain-specific question-answering chatbot for farmer support and agricultural guidance.

SELECTED PROJECTS

- 3D Convective-Cloud Synthesis and Segmentation** [Models] — Developed a 3D diffusion model for convective-cloud synthesis and generated atmospheric volumes with aligned masks; benchmarked eight 3D segmentation architectures, including custom SegFormer3D, using distance, seed, and boundary prediction with watershed-based instance separation.
- Knowledge-Guided Dermatology Image Captioning** [Code] — Built a multimodal RAG pipeline over 16,577 Fitzpatrick17k images, retrieving label-specific DermNet knowledge and extracting key concepts with KeyBERT and BioBERT to ground Qwen2.5-VL captions; created a $\sim 9,800$ -caption clinical dataset, fine-tuned SmolVLM-500M and MedGemma-4B with 4-bit LoRA, and evaluated caption quality and fairness across Fitzpatrick skin-tone groups.

- **CapClipper: Social Media Campaign Analysis** [Code] — Co-developed a tool to detect coordinated promotion on X, applying lexicon- and LLM-based sentiment analysis to GLP-1 posts from a 395M-post corpus. Clustered normalized posts across accounts and filtered out news and organic reposts, surfacing indicators of grey-market peptide promotion.
- **Gastrointestinal VQA and Synthetic Image Generation** [Code] — Fine-tuned Florence-2 on 6,500 gastrointestinal images for visual question answering, achieving 0.92 ROUGE-1 and 0.91 ROUGE-L; adapted Stable Diffusion 2.1 with DreamBooth and LoRA to generate synthetic colonoscopy images.
- **Generative Model Fingerprinting for Synthetic Images** [Paper] — Fused BLIP-2 and DINOv2 embeddings with dimensionality reduction and unsupervised clustering to detect visual fingerprints and attribute synthetic lung CT images to their source GAN or diffusion model generator, achieving a 0.90 Adjusted Rand Index.

HONORS & AWARDS

2026 – First Place, Memetics Track, HopHacks Hackathon, Johns Hopkins University
2026 – HPC Resource Allocation Award (11,249 GPU hours; value \$5,930), Principal Investigator, NSF ACCESS
2025 – Best M.S. Thesis Award, Morgan State University
2024 – IBM Master’s Fellowship Award
2018 – Undergraduate Merit Scholarship, North South University

RECENT PUBLICATIONS

- [1] **M. Hoque** and F. Kim, “Generative modeling of additive manufacturing lack-of-fusion defects using three-dimensional adversarial network,” *Advanced Manufacturing Series (NIST AMS) 100-73*, 2026.
- [2] **M. Hoque** et al., “An Empirical Evaluation of Low-Rank Adapted Vision-Language Models for Radiology Image Captioning,” *Bioengineering*, 2025.
- [3] **M. Hoque** et al., “Comparative Analysis of Fine-Tuned Multimodal Models in Radiology Image Captioning,” *IEEE International Conference on Computing and Machine Intelligence (ICMI)*, 2025.
- [4] Md. I. S. Emon, **M. Hoque**, et al., “A Novel Vision Transformer-Based Approach to Detect Generative Model Fingerprint,” *SPIE Medical Imaging: Imaging Informatics*, 2025.
- [5] E. P. O. Oluwafemi, **M. Hoque**, et al., “Solving Medical Data Limitations Through AI: Multi-Modal Vision-Language Learning for Gastrointestinal VQA and Synthetic Training Data Generation,” *CLEF*, 2025.
- [6] **M. Hoque** et al., “Model-Intrinsic Attention as Explanation for Radiology Image Captioning,” *CLEF*, 2026. **Accepted.**
- [7] R. N. Chowdhury, **M. Hoque**, et al., “Knowledge-Guided Dermatology Image Captioning Using Resource-Efficient Vision-Language Models,” *Computer Methods and Programs in Biomedicine*, 2026. Under review.

PATENTS

- [P1] R. N. Chowdhury, **M. Hoque**, et al., “Method to Generate Dermatology Image Captioning Using Resource-Efficient Vision Language Models,” Morgan State University Office of Technology Transfer, 2026. Patent application submitted.

PROFESSIONAL SERVICE, LEADERSHIP & MENTORSHIP

2025–Present – Graduate Teaching Assistant, Introduction to Data Science, Morgan State University
2025–Present – Peer Reviewer, *IEEE Journal of Biomedical and Health Informatics* and *IEEE Access*
2024–Present – General Secretary, Bangladesh Student Association, Morgan State University
2024–2025 – Executive Board and SCMNS Representative, Graduate Student Association, Morgan State University
2021–2022 – Undergraduate Teaching Assistant, Probability & Statistics and Physics II, North South University

EDUCATION

Morgan State University	Baltimore, MD
<i>M.S. in Advanced Computing (GPA: 3.77/4.00)</i>	<i>Aug. 2023 – May 2025</i>
North South University	Dhaka, Bangladesh
<i>B.S. in Computer Science and Engineering (GPA: 3.73/4.00)</i>	<i>Jan. 2018 – Nov. 2022</i>